MATH 321/STAT 321: Probability Theory
Fall 2026

Instructor Information

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Class Information

Dates: September 14 – November 22
Time: Period 1: 8:00 – 9:10 a.m.
Classroom: Science + Math Center - A203
Website: https://tquinlivan.github.io/

Course Description

An advanced study of probability theory. Sample spaces, random variables and their distributions, conditional probability and independence, transformations of random variables.

Course Objectives

Students completing this course will be able to:

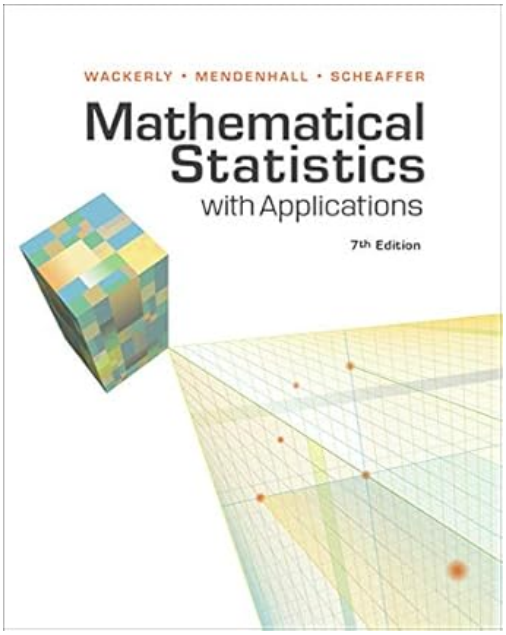
1. Recognize and work with standard models of random phenomena.
2. Understand and express assumptions that underlie random models.
3. Carry out computations in probabilistic models, including simulation.
4. Demonstrate comprehension of the main concepts of probability.

However, this course is the gateway to many different areas of application, and we will include brief introductions to these various topics,including:

- **Actuarial science.** The syllabus for the Society of Actuaries Exam P has an extensive list of Learning Outcomes for what students should learn in probability.
- **Financial Mathematics.** Much of financial mathematics is built on the idea of stochastic processes.
- **Statistics.** Inferential statistics is fundamentally about using data to make estimates about the parameters (such as “mu” and “sigma”) for a probability density functions.
- **Machine Learning.** One of the three main focuses of machine learning is unsupervised learning, which is essentially a collection of techniques for understanding the underlying probability distributions. In addition to parameter estimates for probability density functions, we will also talk about clustering algorithms and dimensionality reduction – two of the main topics in unsupervised learning

Textbook, Calculators, & Software

I will be teaching out of "Mathematical Statistics with Applications" (7th Edition) by Dennis Wackerly, William Mendenhall, Richard Scheaffer. It can be expensive depending on what site you're looking at, but I have seen cheaper listings used if you want a physical copy, and (you didn't hear this from me) but if you search the books title in Google, the second result is from ResearchGate and might just be a full PDF copy of it. While I am going to assign problems out of the textbook, I plan to scan my copy of the problems from the 7th edition, so if you find an older copy of the book, it should be alright to use. The R applications might be out of date or in S or something, but the material should still hold up.



Calculator: You will likely need a calculator to do the computations that will arise throughout the course. No specific calculator is required; I recommend the TI-30XS MultiView (https://a.co/d/0QHmuee). It's cheap and should do anything you need to in this course. You don't need a graphing calculator for this course, but if you have a TI-84 already, it's more than capable.

Class Attendance and Participation

It is essential to your success in this course that you attend and participate in each lecture. I will not take attendance directly, however there is an element of your grade that will include in-class elements. In my experience as a student, a teaching assistant and as a lecturer, the number one most common problem students have in courses is falling behind in the course and not being able to catch back up. If you are forced to miss a day, it is your responsibility to catch yourself back up, whether through getting notes from a classmate or coming to me to go over the material.

In class, I ask that you respect my time, your time and your classmate's time. In class, please avoid anything that would be too distracting your classmate: no cell phones, music, loud foods, etc. I personally recommend taking notes with pen and paper, but if you prefer using a laptop/tablet, please stay focused on the lecture and don't go surfing the web or play *League of Legends*.

I aim to create an open and welcoming environment for learning, so if you have questions or don't understand something I've said, please speak up. If you are confused about something, it's almost certain that you're not the only one, and I'm happy to clarify. If you notice a mistake in my math, let me know, I'm only human and I promise I've made more stupid math mistakes than you have! And if you're curious or want to know more about a subject, speak up, I got my PhD in Mathematics & Statistics because I love this stuff!

Email Policy

If you have any questions outside of class, you can stop by my office, but you can also email me. I will attempt to respond to all emails within 24 hours. If you send an email late at night, I may not respond until the next day, and if you email on the weekends, I may not respond until the following Monday. If I fail to respond after 24 hours during the week, please resend the email, as I may have just not noticed it. I recommend that you email me using your Knox College email account, and keep note that there may be topics I may not be able to discuss over email or feel comfortable discussing over email, in which case I will make a time for us to meet in person to speak.

Academic Integrity & Honor Code

Remember that all work you submit must be in accordance with the Knox College Honor Code (https://www.knox.edu/offices/academic-affairs/honor-code-and-procedures/). Should I find evidence of a violation of academic integrity, I will sanction you.

Note two passages from the Honor Code in particular:

> *Students are expected to leave books and notes elsewhere during an examination and to take examinations in public places such as classrooms or open lounge areas nearby in the buildings. Washrooms, storage areas, and maintenance areas in the basement and elsewhere are not public places. Neither are carrels or other closed areas . . .*
>
> *Faculty may require that students not bring books, notes or electronic devices into the building in which an examination is administered. Alternatively, they may permit students to bring these resources into the building but may require that these books, notes and electronic devices be left in a central area or the classroom throughout the duration of the examination or other academic obligation.*

In accordance with these policies, you should not have a cell phone on you when taking exams or quizzes. I will not allow computers or other electronic devices, notes, textbooks or any other help on exams.

Homework can be done cooperatively, but any submissions must be in your own words.

Using AIs such as Chat-GPT to write your answers and do the work is expressly forbidden, both because they will not help you learn (the only way to learn mathematics is through doing mathematics), and because it doesn't often will not answer fully correctly especially for harder problems/proofs. You will be expected to both fully justify your answers in the homework, and to be able to answer questions in in-class scenarios like quizzes and tests, so while tools like GPT or Google can help to understand concepts or point you in the right direction, you need to internalize the material at some point, and using Chat-GPT to do everything will never give you that chance. If you are struggling with assignments, please *please* ***please*** come speak to me! I don't bite!

Office of Disability Support Services

The Office of Disability Support Services website can be found at https://www.knox.edu/offices/disability-support-services.

Knox College abides by Section 504 of the Rehabilitation Act of 1973 (https://www.epa.gov/external-civil-rights/section-504-rehabilitation-act-1973), which stipulates:

> *No otherwise qualified individual with a disability in the United States, as defined in section 7(20) shall, solely by reason of her or his disability, be excluded from the participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving Federal financial assistance or under any program or activity conducted by any Executive agency or by the United States Postal Service.*

Disabilities covered by law include, but are not limited to, learning disabilities, psychological disabilities, health impairments, hearing, and sight or mobility impairments.

If you have a disability that may have some impact on your work in class and for which you may require accommodations, please contact Patricia Harden in the Office of Disability Support Services (office: SMC E-115; email: pharden@knox.edu) so that such accommodations may be arranged.

Note that the accommodations are not retroactive; I do not make adjustments until ODSS contacts me to let me know what I can do. Also note that accommodations must be renewed each term.

Tutoring

The Center for Teaching and Learning (CTL) offers many resources for students in need of help. Asking for help is no reason to be embarrassed; seeking help in a big part of the learning process, and a sign that you are engaging with the material and your coursework. You are not alone at Knox College and you should utilize all the tools at your disposal to succeed.

Writing Tutors: If you want help on a paper, no matter the course, you can find writing tutors in the Writer's Workshop, which is at the Center for Teaching and Learning (466 S. West St., across from Post Hall), Monday through Friday, from 10:30 to 4:00pm, and again in Red Room on Tuesday, Wednesday, and Thursday nights from 7:00 to 9:00pm. You can also schedule appointments to meet with a writing tutor either online or in person (should you both so desire) by visiting the Center for Teaching and Learning website.

There is no tutor attached to this class, and I do not know if any Red Room tutors there have taken this class before, so it may be difficult finding help there for this class. I recommend starting a group with your fellow classmates to study together, and as I've said repeatedly here, come talk to me in office hours early and often!

TimelyCare

If you're unfamiliar with mental health support, taking care of your mental well-being can be stressful, or even scary. Knox is partnered with TimelyCare to provide a virtual health care solution that reflects Knox's values and student population. It is a student-centered platform that offers on-demand access to mental healthcare and a diverse, culturally competent provider network. TimelyCare complements Knox's on-campus services with 24/7 on-demand virtual mental health care, health coaching, peer to peer support and a plethora of coping strategies ranging from homesickness to depression to study skills. There are 12 free sessions available from September to August annually and can be accessed anywhere on or off campus (even internationally!). To access TimelyCare, go to TimelyCare.com or use the QR code below, register with your Knox email and follow prompts! If you have any questions, please reach out to Knox Health and Wellness at counseling@knox.edu or (309) 341-7492. If there is an emergency, please call 911 or contact Campus Safety at (309) 341-7979.



Title IX

Knox College values a culture of respect. If you have encountered any form of discrimination or harassment, you are encouraged to report it at https://www.knox.edu/about-knox/our-values/culture-of-respect/report-it.

As a faculty member, I am required to disclose to the Director of Title IX & Civil Rights Compliance any information I receive about discrimination, harassment, sexual misconduct and other forms of interpersonal violence. This includes information shared in class assignments, discussions, emails, and/or shared conversations outside of class. If a disclosure is received, she will reach out to you. You are not required to respond to this outreach.

If you wish to speak with a confidential resource (one not required to report to Title IX), please contact:

- Knox College Health Services (https://www.knox.edu/offices/health-and-counseling-services/health-services)
- Knox College Counseling Services (https://www.knox.edu/offices/health-and-counseling-services/counseling-services)
- Knox College Director of Spiritual Life (https://www.knox.edu/offices/spiritual-life/meet-the-staff)
- TimelyCare (https://app.timelycare.com/auth/login)

Pregnancy and/or Pregnancy Related Conditions

Faculty members are required to inform a student who is pregnant or experiencing a pregnancy related condition (including termination) that the Director of Title IX & Civil Rights Compliance is available to provide support by helping you to understand your rights/options and provide supportive measures. You can contact the Director of Title IX & Civil Rights Compliance at cultureofrespect@knox.edu.

Grading Information

Exams

We will have two midterm exams in this course, tentatively in the 4th and the 8th week of the course. The exact date will be confirmed as we approach the date. Each test will account for about **15% of your grade**, for a total of **30% of your grade**.

At the end of the course, we will have a cumulative final exam during the time scheduled by the registrar's office. It will count for about **20% of your grade**.

Exams will be given in the classroom that we do lectures in, and I will expect all of you to leave your cell phones at the front of the room when you take it.

If you must miss a quiz or exam, you will need to talk to me ahead of time, as soon as you realize it, or you must have a good reason if you come to me after the fact.

Homework Assignments

Homework will be done as a combination of textbook problems and other questions, to be answered by hand (or if you want to be especially studious, you may use LaTeX!), though some may involve computations/programming. The assignments will total to **25% of your grade**. I expect the homework answers to be fully answered with all necessary work and justification shown, and to be in your own words. Also, make it legible please! I can't grade what I can't read. Don't be afraid to use more than one sheet of paper, I promise I'll recycle everything in the end! Like I used to say as a TA, come to my office, I will give you looseleaf paper if you need it! It's not that expensive.

For each day that an assignment is late, I will take 20% off the grade.

Quizzes

On Thursdays that are not exam days, I will set aside some time for any questions you have on the weeks material, and then I will give a quiz or assessment of some kind. These will amount to a total of **20% of your grade**.

If you must miss a quiz or exam, you will need to talk to me ahead of time, as soon as you realize it, or you must have a good reason if you come to me after the fact. All quizzes must be made up within 24 hours, or I will take the points that would have gone to that quiz and redistribute them onto the other quizzes.

Attendance

During most lectures, I will try to assign small assignments, likely a practice problem done together or a small "quiz", or simply take attendance. These will total to **5% of your grade** overall. I will drop enough of these that you will have a few days to be able to miss, but I urge you in the strongest possible way to please *please* attend all lectures.

Grading Breakdown

Thus the grades will break down as follows:

Component	Points
Midterm Exam #1	15%
Midterm Exam #2	15%
Final Exam	20%
Homework	25%
Quizzes	20%
Attendance	5%

Grade Scale

Final grades will be assigned according to the following percentages, though I reserve the right to alter the ranges as need be:

Grade	Range
A	>93%
A-	90% – 93%
B+	87% – 90%
B	83% – 87%
B-	80% – 83%
C+	77% – 80%
C	73% – 77%
C-	70% – 73%
D+	67% – 70%
D	63% – 67%
D-	60% – 63%
F	<60%